Training Dynamics of Overparameterized Neural Networks

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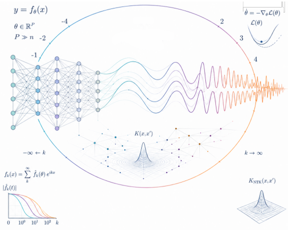

Spectral Bias

NTK on S^1

Fourier Analysis

Finite-Scale Bounds

Supervised Learning: Fitting data to a function

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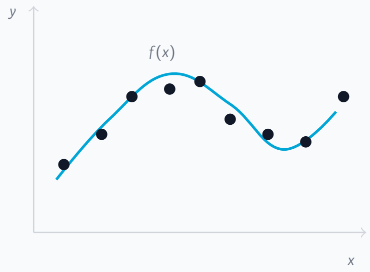

Supervised Learning: Fitting data to a function

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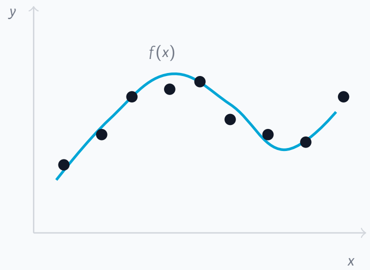
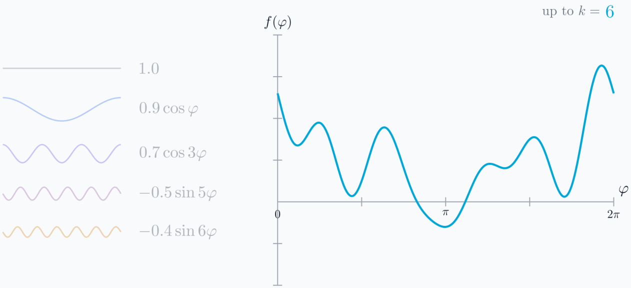
How: choose f so its predictions are close to the target function on the observed data, i.e. minimise $\frac{1}{n}\sum_{i=1}^{n}\left(f(x_i) - f^\star(x_i)\right)^2$.



Example: a concrete target

A function is built from a coarse shape plus finer and finer detail.



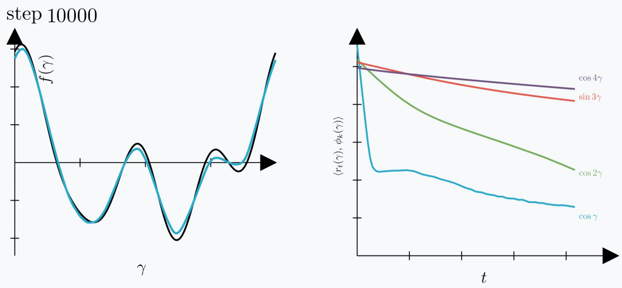
Frequency components

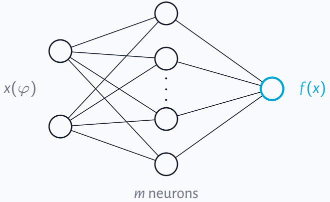
- Each piece is one frequency, from slow waves to fast ones.
- Low frequencies set the coarse shape; high frequencies add the fine detail.
- The target is a short sum of these pieces.

How do we learn this function?

Watch one network learn it

Train a single-hidden-layer network on that target and watch the residual — what gets learned first?





$$f_m(x;\theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} a_i\, \sigma\left(w_i^\top x + b_i\right), \quad \sigma = \mathrm{ReLU}$$

The coarse shape appears first; the fine detail comes last.

What to watch

- The fit climbs toward the target.
- Low-frequency modes die first.
- Fine detail fills in last.

Why does gradient descent learn simple structure first?

A spectral view of training dynamics

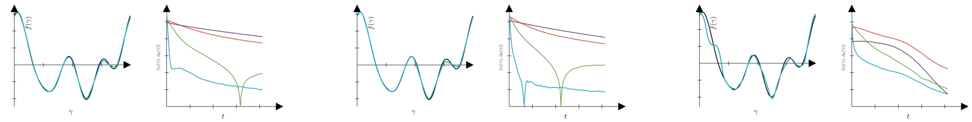
Thesis in one sentence

Reformulate spectral bias as an operator eigenvalue problem, then quantify how the picture changes at finite sample size and finite network width.

Three questions

1. What controls mode-wise learning rates?
2. Does the explanation survive finite data?
3. Does it survive finite width?

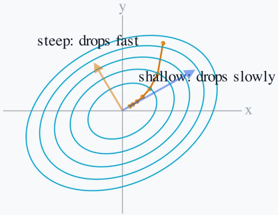
Low-frequency structure is learned before high-frequency detail.



One-hidden-layer MLP — Two-hidden-layer MLP — Attention Block

The error is a landscape

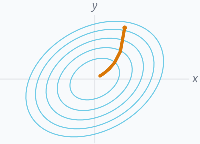
Seen from above, the contour rings are the bowl's height; training rolls toward the bottom.



Each direction has its own rate

Training makes the error small by walking downhill, along the steepest drop.

recall the error: $\frac{1}{n}\sum_i \left(f(x_i) - f^\star(x_i)\right)^2$



In your x, y axes the route bends: the two directions are coupled.

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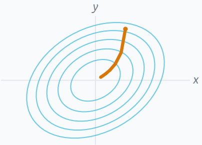

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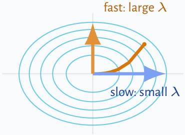

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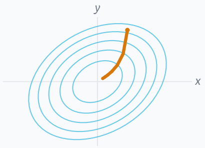

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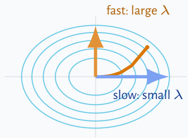

Along each axis the error falls on its own.

Along one direction, the error falls in proportion to how much is left:

$$\dot{\alpha}(t) = -\lambda\,\alpha(t)$$

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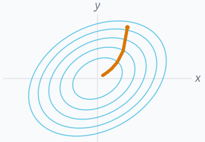

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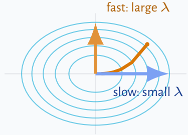

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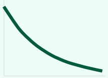

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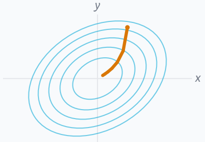

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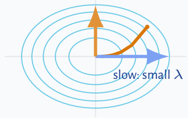

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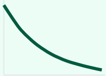

like cooling or half-life; λ is the rate.

Eigenvector: a direction the dynamics keeps separate, never mixing it with others. **Eigenvalue** λ: its rate, large is fast, small is slow.

For a neural network, what are these directions and rates?

The rate becomes an operator: the tangent kernel

The same law as before, with the single rate replaced by an operator.

One direction

$$\dot{\alpha}(t) = -\lambda\,\alpha(t)$$

The error falls at rate λ, a single number.

A whole function

$$\partial_t r_t = -T_t\, r_t$$

The rate becomes an operator T_t, the network's tangent kernel (NTK).

If T_t were fixed, diagonalizing it would split this into one copy of the scalar law per eigenfunction, $\dot{\alpha}_k = -\lambda_k \alpha_k$. The eigenfunctions are the directions; the eigenvalues are the rates.

So the question becomes: what is the spectrum of T?

Why we start from an idealized model

To read off a fixed set of directions and rates, T must be one fixed operator. Two things are in the way.

Time-dependent

T_t changes as the network trains, so its directions and rates drift.

Freeze it to get a constant operator.

Finite scale

At finite data T depends on the sampled points; at finite width it is random.

Take the continuum and infinite width for one deterministic operator.

Strategy: build the clean reference, then put each idealization back.

Reference model ⟶ Finite data ⟶ Finite width ⟶ Evolving

SETUP

Setting: regression on S^1

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$$x(\varphi) = (\cos\varphi, \sin\varphi) \in S^1$$

$$y(\varphi) = \sum_k c_k \phi_k(\varphi)$$

Fourier basis: $1, \cos(k\varphi), \sin(k\varphi)$.

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Key question: what governs the frequency-dependent decay of $\alpha_k(t) = \langle r_t, \phi_k \rangle$?

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$$\lambda_k = \begin{cases} \frac{1}{4} + \frac{3}{\pi^2}, & k = 0, \\ \frac{1}{4} + \frac{1}{\pi^2}, & k = 1, \\ \frac{k^2+3}{\pi^2(k^2-1)^2}, & k \geq 2 \text{ even}, \\ \frac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd}. \end{cases}$$

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Reference model: exact Fourier dynamics at infinite width and in the continuum.

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The ideal model is useful only if we understand its perturbations.

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Finite samples

$$\mu \quad \rightsquigarrow \quad \frac{1}{n}\sum_{i=1}^{n}\delta_{\varphi_i}$$

Replace the continuous measure by n random points on S^1.

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$$T_\infty \quad \rightsquigarrow \quad T_m^{(0)}$$

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Question

Does the finite-width tangent operator preserve the Fourier eigenspaces and learning rates?

Both perturbations break exact diagonalization; the goal is to quantify the finite-scale error.

NON-ASYMPTOTIC CONTROL

Main results: explicit finite-scale error bounds

On a fixed low-frequency Fourier subspace $\mathcal{H}_K$ with $d = 2K + 1$, both perturbations are controlled.

Finite samples

On the Fourier block $\mathcal{H}_K$:

$$A_{ij} = \frac{1}{n} T_\infty(\theta_i, \theta_j)$$

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$$\Lambda_K = P_K T_\infty P_K$$

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Finite samples

Thm. 5.5 + Cor. 5.6

With probability at least $1 - \delta$:

$$\|G - I\|_{\max} \leq C_1 \sqrt{\frac{\log(d/\delta)}{n}},$$

$$\|H - \Lambda^{(n)}\|_{\max} \leq C_2 \sqrt{\frac{\log(d/\delta)}{n}},$$

$$\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \leq C_3 \sqrt{\frac{\log(nd/\delta)}{n}}.$$

Frozen finite width

Thm. 5.7

With probability at least $1 - \delta$:

$$\mathbb{E}\, B_m^{(K)} = \Lambda_K$$

$$\|B_m^{(K)} - \Lambda_K\|_{\max} \leq C_4 \sqrt{\frac{\log(d/\delta)}{m}}$$

Summary: Fourier structure survives finite sampling and frozen finite width with explicit high-probability control.

A finite-width effect not explained by the frozen NTK

After controlling the frozen operator, the next question is whether training changes the operator enough to matter.

EVOLVING KERNEL

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

A finite-width effect not explained by the frozen NTK

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

[Figure] Training loss: frozen vs. evolving kernel — width $m = 1024$

width $m = 1024$: frozen picture becomes competitive

- At small width, the evolving network is not well described by simply freezing the NTK at initialization.
- As width grows, the difference between frozen and evolving dynamics becomes much smaller.
- This is the main tension: at small width the kernel moves enough to matter, so the frozen approximation no longer describes training, even though the low-to-high ordering is unchanged.

Why it helps: added strength, not better alignment

At small width the kernel reshapes itself, but not where the frozen theory would look.

[Figure] Fourier alignment over training (subspace cosine similarity), per frequency

Alignment with the Fourier modes does not improve, and degrades at higher frequencies.

[Figure] Low-frequency spectral strength over training, per frequency block

Low-frequency spectral strength rises sharply, mostly in $k = 0$ and $k = 1$, and it tracks the lower loss.

Spectral bias is reinforced. What breaks is the frozen description, not the low-to-high ordering.

- Shown as a correlation with lower loss, not an established cause.
- A small-width effect: at large width the frozen picture wins (the $m = 1024$ reversal), as lazy training predicts.

CONCLUSION

What we can now say

The three questions from the start, answered.

1 Rates

Mode-wise rates are eigenvalues. Low frequencies have larger eigenvalues and are learned first; $\lambda_k \sim k^{-2}$.

2 Finite data

Finite samples and frozen finite width preserve the picture, with explicit $O(n^{-1/2})$ and $O(m^{-1/2})$ control.

3 Finite width

Training evolves the kernel and reinforces the low-frequency bias. The frozen description holds where the kernel stays lazy, at large width.

SCOPE

What is special about the circle, and what is not

Special

Uniform measure on S^1 makes the NTK a convolution, so its eigenfunctions are exactly the Fourier modes.

Carries over

Off this setting (non-uniform density, or Gaussian inputs giving Hermite polynomials) the basis changes, but eigenvalue size still sets the rates.

Open problem

A non-asymptotic theory for the evolving kernel: control of $T_t^{(m)} - T_0^{(m)}$ over training time.

Spectral bias is an eigenvalue statement: rate equals eigenvalue.

It survives finite data and finite width, and at small width training reshapes the kernel to favour low frequencies even more.

Questions?

[acknowledgements placeholder]

Backup

NTK: from parameter updates to an operator on functions

[Backup] Derivation of the NTK: how gradient descent on weights becomes $\partial_t r_t = -T_t r_t$

Concentration bounds: the constants, and where the magnitude bound is loose

[Backup] Explicit constants in the $O(n^{-1/2})$ and $O(m^{-1/2})$ bounds; discussion of where the magnitude bound is not tight

Computing the eigenvalues λ_k

[Backup] Integral computation of λ_k for the ReLU NTK on S^1; verification against the explicit cases $k = 0, 1$

The $k = 2$ anomaly

[Backup] Why λ_2 does not follow the generic even-k formula; how the $k^2 - 1$ denominator is modified at $k = 2$

Strict ordering: where it holds and where it does not

[Backup] Conditions under which $\lambda_k > \lambda_{k+1}$; counterexamples or caveats near the transition between even and odd cases

Why $k = 0$ and $k = 1$ gain strength: a hypothesis and its test

[Backup] Parameterisation hypothesis; proposed clean test (target with no $k = 0, 1$ component)

Mean-field and preconditioning views

[Backup] Mean-field interpretation of infinite-width limit; preconditioning perspective on spectral bias